

13. Array Three way partitioning of an array around a given value

```
class Solution{
```

```
public:
```

```
void threeWayPartition(vector<int>& arr, int a, int b)
```

```
{
```

```
    int low = 0, mid = 0, high = arr.size() - 1;
```

```
    while(mid <= high) {
```

```
        if(arr[mid] < a) {
```

```
            swap(arr[low], arr[mid]);
```

```
            low++;
```

```
            mid++;
```

```
        }
```

```
        else if(arr[mid] > b) {
```

```
            swap(arr[mid], arr[high]);
```

```
            high--;
```

```
        }
```

```
        else {
```

```
            mid++;
```

```
        }
```

```
    }
```

```
}
```

```
};
```

The screenshot displays a coding platform interface with a dark theme. On the left, the 'Output Window' shows 'Compilation Results' for 'Custom Input' by 'Y.O.G.I. (AI Bot)'. It confirms 'Problem Solved Successfully' with 1111/1111 test cases passed, 1/1 attempts, 100% accuracy, 2/2 points scored, and a time taken of 0.03 seconds. Below this, 'Solve Next' buttons are visible for 'Wave Array', 'Sort by Absolute Difference', and 'Convert an array to reduced form'. On the right, the C++ code for the 'threeWayPartition' function is shown, matching the code provided in the text above. The code uses a while loop to partition the array around two values, 'a' and 'b', by moving elements to the low, mid, or high indices.

```
1 class Solution{
2 public:
3     void threeWayPartition(vector<int>& arr, int a, int b)
4     {
5         int low = 0, mid = 0, high = arr.size() - 1;
6
7         while(mid <= high) {
8             if(arr[mid] < a) {
9                 swap(arr[low], arr[mid]);
10                low++;
11                mid++;
12            }
13            else if(arr[mid] > b) {
14                swap(arr[mid], arr[high]);
15                high--;
16            }
17            else {
18                mid++;
19            }
20        }
21    }
22 };
23
```